INCH-POUND

MIL-DTL-32025(OS)
11 June 1998

DETAIL SPECIFICATION

CARTRIDGE, IMPULSE, CCU-96/B

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the manufacture and acceptance of the CCU-96/B Impulse Cartridge.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or documents recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements for documents referenced in sections 3 and 4 of this specification, whether or not those documents are listed in this section.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices
MIL-I-23659	Initiators, Electrical, General Design Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Technical Information Division (Code 840), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-810	Test Method Standard for Environmental Engineering Considerations and Laboratory Tests
DOD-STD-2101	Classification of Characteristics

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

DL1762AS106	Cartridge, Impulse, CCU-96/B
1762AS106	Cartridge, Impulse, CCU-96/B
1762AS100	Cutter, Cartridge Actuated, MLU-57/B
2842665	Wire, TOW

(Application for copies should be addressed to Commanding Officer, Indian Head Division, Naval Surface Warfare Center, Data Control Team (Code 8410P), Indian Head, MD 20640-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR QUALITY CONTROL

ANSI/ASQC Z1.4	Sampling Procedures and Tables for Inspection by Attributes

(Application for copies should be addressed to the American Society for Quality Control, PO Box 3005, 611 E. Wisconsin Ave., Milwaukee, WI 53201-4606.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 1742	Standard Practice for Radiographic Examination

(Application for copies of ASTMs should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

INSTITUTE FOR INTERCONNECTING AND PACKAGING ELECTRONIC CIRCUITS

J-STD-006	Requirements for Electronic Grade Solder Alloys and Fluxed and Non-Fluxed Solid Solders for Electronic Soldering Applications

(Application for copies should be addressed to IPC, 7380 N. Lincoln Avenue, Lincolnwood, IL 60646.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection (see 4.3).

3.2 Conformance to documents. The CCU-96/B Impulse Cartridge, 1762AS106, covered by this specification shall be manufactured in accordance with DL1762AS106 and all documents listed thereon.

3.3 Cartridge inspection.

3.3.1 Cartridge assembly leakage. The cartridge assembly shall not leak when tested in accordance with 4.5.1.

3.3.2 Bridgewire circuit resistance. The resistance of the bridge circuit in each completely assembled cartridge, when measured electrically (see 4.5.2), shall be not less than 0.9 ohm nor greater than 1.1 ohm.

3.3.3 X-ray. Cartridges shall be free of imperfections when x-rayed (see 4.5.3).

3.4 Power-current. The cartridge shall not fire within 5 minutes when subjected to a current of 1 ampere minimum per bridge with an associated power of 1 watt minimum per bridge. The cartridge shall meet this requirement at 70° ± 5°F. The test method shall be as explained in 4.5.4. Follow test routine in table I.

3.5 Ballistic requirements. Cartridges shall meet the ballistic requirements specified in 3.5.1 through 3.5.3 when subjected to the test prescribed in 4.5.5.

3.5.1 Output. Cartridges shall furnish sufficient power to completely sever the steel cable specified in 4.5.5.

3.5.2 Misfire. There shall be no misfires (see 4.5.5.2).

3.5.3 Hangfire. There shall be no hangfires (see 4.5.5.3).

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3.6 Environmental.

3.6.1 Vibration. The cartridges shall withstand vibration conditions as specified in 4.6.1. After being subjected to the vibration test, the cartridges shall meet the nondestructive test requirements and the performance requirements in table I.

3.6.2 Forty-foot drop. To ensure safety of handling, the cartridege shall not fire when dropped from a height of 40 feet as specified in 4.6.2 and shall be safe for handling and disposal. Follow test routine in table I.

3.6.3 Temperature/humidity/altitude cycling. The cartridges shall withstand the temperature/humidity/altitude cycling conditions as specified in 4.6.3. After being subjected to the temperature/humidity/altitude test, the cartridges shall meet the nondestructive test requirements and the performance requirements in table I.

3.7 Workmanship. Cartridges shall be free of the following defects: burrs, dents, missing or damaged O-ring, or other defects which may prevent entry of the cartridge into the firing chamber, or affect the resistance of the cartridge to moisture. All cartridge marking and identification shall be verified as proper.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed as specified in the applicable test method of 4.5.

4.3 First article inspection. First article inspection shall be performed prior to production by a facility designated in the contract or order (see 6.2) and all parts and materials, including packaging and packing, shall be the same as used for regular production and shall be from the intended sources of supply. First article inspection shall be performed on sample units which have been produced using the same materials, equipment, processes, and procedures that will be used in regular production.

4.3.1 Sample size. The first article sample shall consist of 76 cartridges.

4.3.2 Inspection routine. Seventy-six cartridges shall be expended in the tests listed below and 10 shall be retained for investigative purposes. Testing shall consist of the tests specified in table I.

4.3.3 Failure. Failure of any cartridge to meet the requirements of 3.1 through 3.7 shall be cause for rejection of the first article.

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TABLE I. First article inspection.

Nature of test	Requirement	Sample size						
Visual Exam	3.7	6	12	12	12	12	12	10
X-Ray	3.3.3	6	12	12	12	12	12	10
Resistance	3.3.2	6	12	12	12	12	12	10
Leakage	3.3.1	6	12	12	12	12	12	10
Power current	3.4	6	12	12	12	12	12	10
Resistance	3.3.2	6	12	12	12	12	12	10
40 foot drop	3.6.2	6						
Vibration	3.6.1			12				
X-Ray	3.3.3	6		12				
Temp./Hum./Alt.	3.6.3		12					
Resistance	3.3.2		12	12				
Fire at:	3.5							
-65 °F					12			
+70 °F			12	12		12		
+160 °F							12	
Discards		6						
Retains								10

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the tests specified in table II and verification of the characteristics classified on the drawings of DL1762AS106.

TABLE II. Quality conformance tests and examinations.

Item	Nature of test	Requirement	Test Method	Classification (DOD-STD-2101)
Cartridge	Leakage	3.3.1	4.5.1	(C1)
Cartridge	Bridgewire circuit resistance	3.3.2	4.5.2	(C2)
Cartridge	X-ray	3.3.3	4.5.3	(C3)
Cartridge	Ballistics	3.5	4.5.5	(C4)
Cartridge	Visual inspection	3.7		

4.4.1 Lot size. Unless otherwise specified (see 6.2), the production lot size shall be not less than 151 and not greater than 500. Only primary components from a single lot shall be used in a production lot of cartridges.

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4.4.2 Sampling. Unless otherwise specified (see 6.2), sampling shall be in accordance with table III of this specification.

TABLE III. Quality conformance sample size.

Lot Size	Sample size	Retained sample
01-50	8	2
51 - 90	13	2
91 - 150	20	2
151 - 280	32	2
281 - 500	50	2
501 - 1200	80	2

4.4.3 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as critical, major, or minor in accordance with DOD-STD-2101. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

4.5 Tests.

4.5.1 Leakage. Each cartridge shall be leak tested in a dry gas leak tester. Cartridges which exhibit a leak test rate in excess of 10^{-5} cm^3/s gas or air at a pressure differential of 1.0 ± 0.1 atmosphere shall be considered defective and rejected.

4.5.2 Bridgewire circuit resistance. Conformance with the bridge circuit resistance of 3.3.2 shall be ascertained on all completely assembled cartridges by means of a test circuit which limits the bridge circuit current to 25 milliamperes, maximum. A suitable instrument which is known to be accurate to within 1 percent in the range of resistance specified shall be used to determine whether or not the requirement is being met. Units which fail to meet the requirements of 3.3.2 shall be removed from the lot.

4.5.3 Radiographic examination. All cartridges shall be radiographically examined in accordance with ASTM E 1742. Cartridges having any observable imperfections in assembly shall be cause for rejection. The cartridges shall be positioned on their sides for the most revealing exposure. All cartridges shall be identified with serial numbers prior to examination. The cartridge serial numbers shall be in consecutive order beginning with the number 001 in each production lot. The cartridges shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the cartridges displayed thereon. The radiograph identification shall include the complete lot number, as stamped on the cartridges, and the span of serial numbers displayed. Discontinuities in serial numbers shall be identified on the x-ray. Defective cartridges found by the vendor shall be clearly identified on the x-ray and those defectives removed from the production lot. The serial numbers shall be pen or ink stamped on the cartridges with approximately 0.118-inch high characters. The serial numbers shall be located apart from the other cartridge markings.

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4.5.4 Power-current. The cartridge shall be preconditioned at 70 ± 5 °F for a period of twelve hours minimum. Perform test in accordance with MIL-I-23659, paragraph 4.4.3.1.

4.5.5 Ballistic test. The cartridges shall be conditioned not less than 4 hours at 70 ± 5 °F. Production samples shall be electrically initiated with a current of 10 ± 0.5 amperes in a cutter MLU-57/B (Drawing 1762AS100). The samples selected shall sever a 3/4-inch-diameter 6 x 19 IWRC wire rope in accordance with drawing 2842664. The cable cutter shall be fired in a horizontal position against the wire rope preloaded with a 200 ± 10 lb tensile load.

4.5.5.1 Criteria of acceptability. The failure of any cartridge to completely sever the steel cable shall be cause for rejection of the lot represented. There shall be no misfires (4.5.5.2) or hangfires (4.5.5.3).

4.5.5.2 Misfire. The failure of any cartridge to fire shall be cause for rejection of the lot represented, unless such failure is plainly attributable to faulty test equipment or procedure.

4.5.5.3 Hangfire. One or more hangfires shall be cause for rejection of the lot represented. A hangfire is defined as a firing in which the elapsed time between actuation of the firing mechanism and complete severance of the steel cable exceeds 1.000 second.

4.6 Environmental

4.6.1 Vibration test. The test shall be conducted with all 12 cartridges installed in the test fixture. The testing shall be 1 hour in the longitudinal axis per temperature and 2 hours in the transverse axis per temperature. All 12 cartridges shall be subjected to all three temperature environments, -65, +70, and +200 °F. The total vibration time will be 9 hours, 3 hours at each temperature. Due to the symmetrical design of the cartridge, testing will only be done in 2 axes, with the 3 hour test time divided, as above, between the two axes. The test will consist of the following two sections:

a. Fixed Dwells:

F1= 20.9Hz	A1= 3.125g
F2= 146.3Hz	A2= 2.25g
F3= 292.6Hz	A3= 2.25g
F4= 438.9Hz	A4= 2.25g

The duration of each fixed dwell shall be 7.5 minutes in the longitudinal axis at each temperature. The duration will be 1.5 minutes in the transverse direction.

b. Cycling:

The remainder of the test time shall be for cycling. Per the original testing of the CCU-96/B Impulse Cartridge. A cycle shall be from 5 to 500 Hz with a 7.5 minute sweep, with a maximum acceleration of 10 g's.

4.6.2 Forty-foot drop test. This test shall be conducted in accordance with test A3 of MIL-STD-331. Six cartridges shall be dropped to impact in the following positions: (1) two with head up, (2) Two with head down, and (3) two horizontal. A new cartridge shall be used for each drop. The cartridges shall meet the requirement of 3.6.2.

4.6.3 Temperature/humidity/altitude cycling. The cartridges shall be subjected to the temperature/ humidity/altitude cycling test as specified in MIL-D-21625 paragraph 4.8.4.

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5. PACKAGING

5.1 Packaging. The packaging requirements shall be as specified in the contract or order (see 6.2 and 6.3). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The CCU-96/B Impulse Cartridge is intended to actuate a military helicopter-rescue hoist cable cutter (MLU-57/B) to jettison a load in an emergency. It is used only in the MLU-57/B and therefore has no commercial application.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2, and 2.3).

c. Whether first article inspection is required, and if so, specify the test activity (see 3.1, and 4.3)

d. Date and place of delivery

e. Inspection conditions, if other than as specified (see 4.2)

f. Lot size, if other than as specified (see 4.4.1)

g. Sampling, if other than as specified (see 4.4.2)

h. Packaging and marking as specified in the contract (see 5.1, and 6.3).

i. That the safety precaution requirements of the "Contractors' Safety Manual for Ammunition and Explosives," DOD 4145.26M, are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore," OP 5 are applicable.

6.3 Packaging. Level A packaging should be used for packaging of all production lots for service use (see 6.2). One cartridge should be tightly packaged in a hermetically sealed can which meets the requirements of MIL-C-10464, type I. Adequate support and cushioning should be provided for the packaged item.

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6.5 Subject term (keyword) listing.

Helicopter
cable cutter

Preparing activity:
Navy - OS
(Project 1377-0085)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-32025 (OS)	**2. DOCUMENT DATE (YYMMDD)** 980611

CARTRIDGE, IMPULSE, CCU-96/B

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7. DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *Include Area Code)*	
COMMANDER, INDIAN HEAD DIVISION	(1) Commercial (301) 743-4700	(2) AUTOVON 354-4700
NAVAL SURFACE WARFARE CENTER (CODE 840M) 101 STRAUSS AVENUE INDIAN HEAD, MD 20640-5035		

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